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ETE 3351

Lab 8

Sum and Difference, Sum and Difference, Sum and Difference, Sum and Difference, Sum and Difference, Sum and Difference, did I mention Sum and Difference

Introduction:

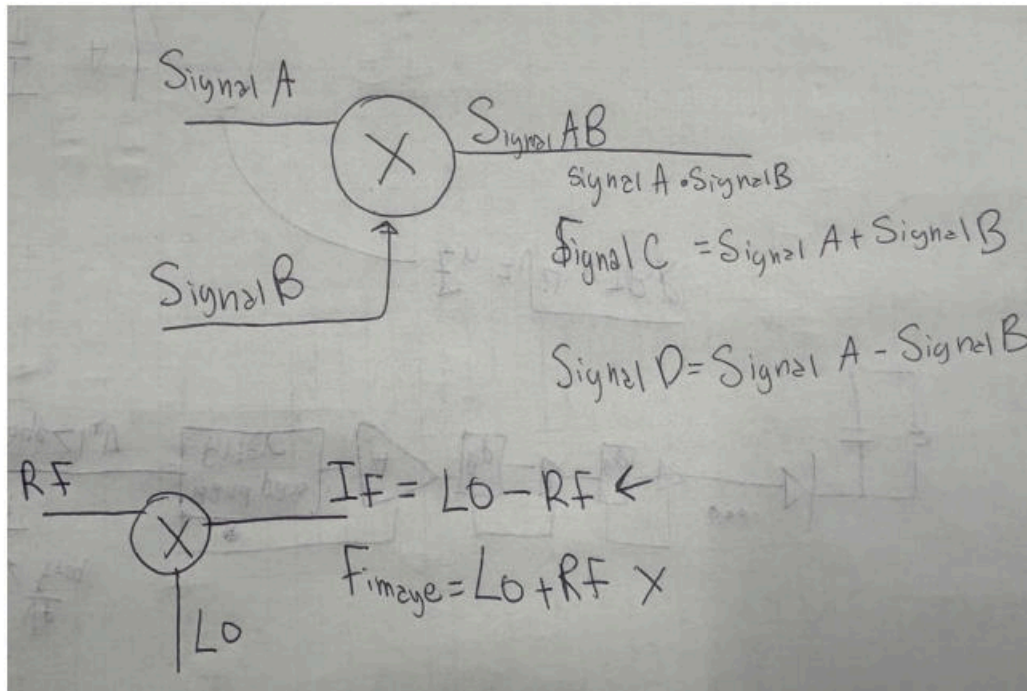
The objective of this lab is to understand the operation of an RF mixer, to perform basic frequency domain tests with the mixer, to perform basic amplitude level tests with the mixer, to understand the effect of adding an IF filter to a mixer and to review typical specifications for a mixer. Mixers and the operation of "mixing" (called heterodyning) are used with the goal to multiply two signals and thus allow mixers to perform frequency conversion. Essentially, mixers are used to overcome the issue of needing a filter to get each desired output frequency.

We use bandpass filters to take advantage of how their property of letting all frequencies in a range pass. Filters are critical for many reasons, but there is a different filter needed for eliminating noise for each desired output frequency. Note that according to the famous noise voltage equation, bandwidth of the bandpass filter is the only part we can actually adjust from this equation.

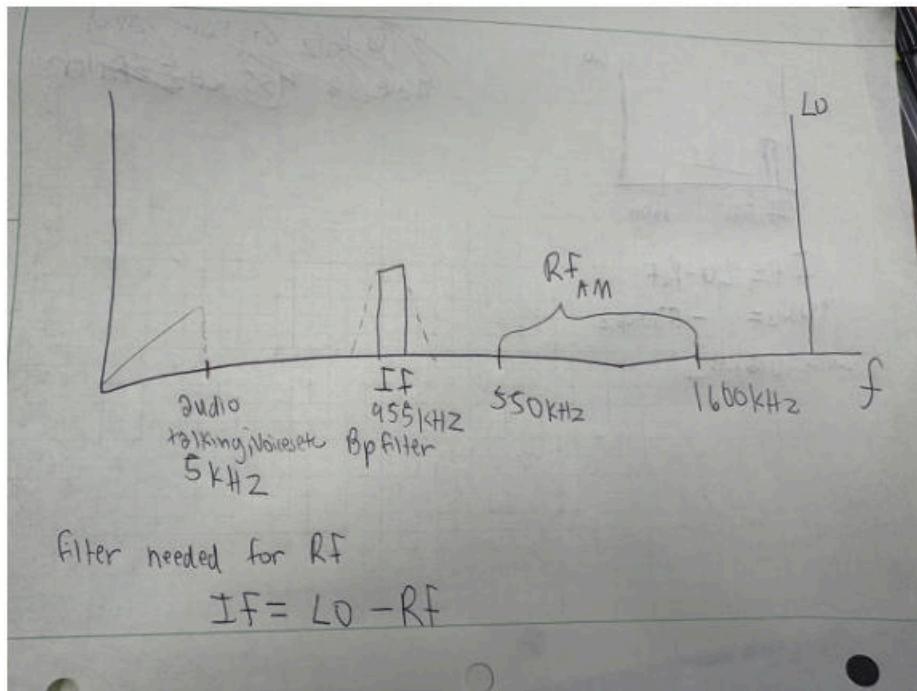
$$E_n = \sqrt{4KTBR}$$

- K = constant value
- T = temperature in Kelvins
- B is the bandwidth
- R is the resistance

Since the needed bandwidth changes based on the desired frequency, we would therefore need a different filter for each frequency in the AM range (550kHz to 1600kHz). Having so many filters for each frequency is NOT ideal, and an adjustable filter is expensive (not to mention, adjustable capacitors are unrealistic). The solution to this conundrum is to use a mixer.



IMG 1. RF is the radio frequency, LO is the local oscillator frequency (and is adjustable)



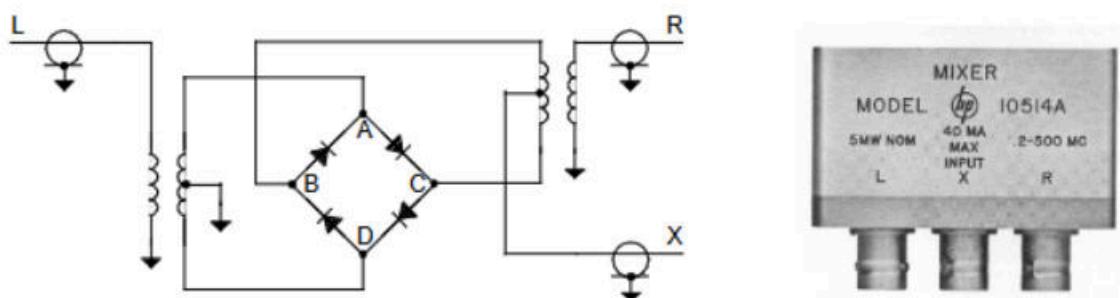
IMG 2.

The IF frequency never changes (455kHz) and therefore only one filter for this frequency is needed. The frequency of the oscillator now is the component that changes, meaning that we do not need an adjustable filter any more. For example, if we want to listen to a 1000kHz station

(RF), we would determine the frequency to set the local oscillator by doing $455\text{kHz} = \text{LO} - 1000\text{kHz}$, and the local oscillator would be 1455kHz .

We can see that the mixer is able to create a sum (Fimage) and difference (IF) frequency-mixers unfortunately have the effect of creating a lot of extra, unwanted frequencies in addition to the fundamentals thanks to the presence of harmonics.

Important: The mixers used in this lab are a combination of RF diodes and transformer coils (see circuit below) and as such can be damaged by excessive currents as there are no internal current limiting resistors (which would cause excessive signal loss) so please monitor and know the voltage level of all input (L and R) generator signals at all times. In reviewing the mixer datasheet, attached to the end of this handout, the maximum input signal rating is 40mA (also engraved on the unit) which corresponds to 2Vpk in a 50 system. Also from the datasheet the recommended operational level for the L input is 5mW which is 1.4Vpk-pk for sine waves (0.5VRMS) in a 50 system, when using the Spectrum analyzer $5\text{mW} \Rightarrow +7\text{dBm}$ (6.989dBm) from $\text{dBm} = 10\log(P/1\text{mW}) = 10\log(5\text{mW}/1\text{mW}) = 10\log(5) = +7\text{dBm}$. The R input will be operated at a level of 1mW or 0dBm which corresponds to 0.63Vpk-pk (630mVpk-pk) in a 50 system



As shown the mixer is a three terminal coaxial device with L, R and X terminals.

The L stands for the Local Oscillator or LO connection, which is normally a variable frequency oscillator.
The LO is what you change when you "tune" your radio.

The R stands for Radio Frequency connection, which is normally the signal desired to be received or the radio station frequency signal you want to hear on your radio.

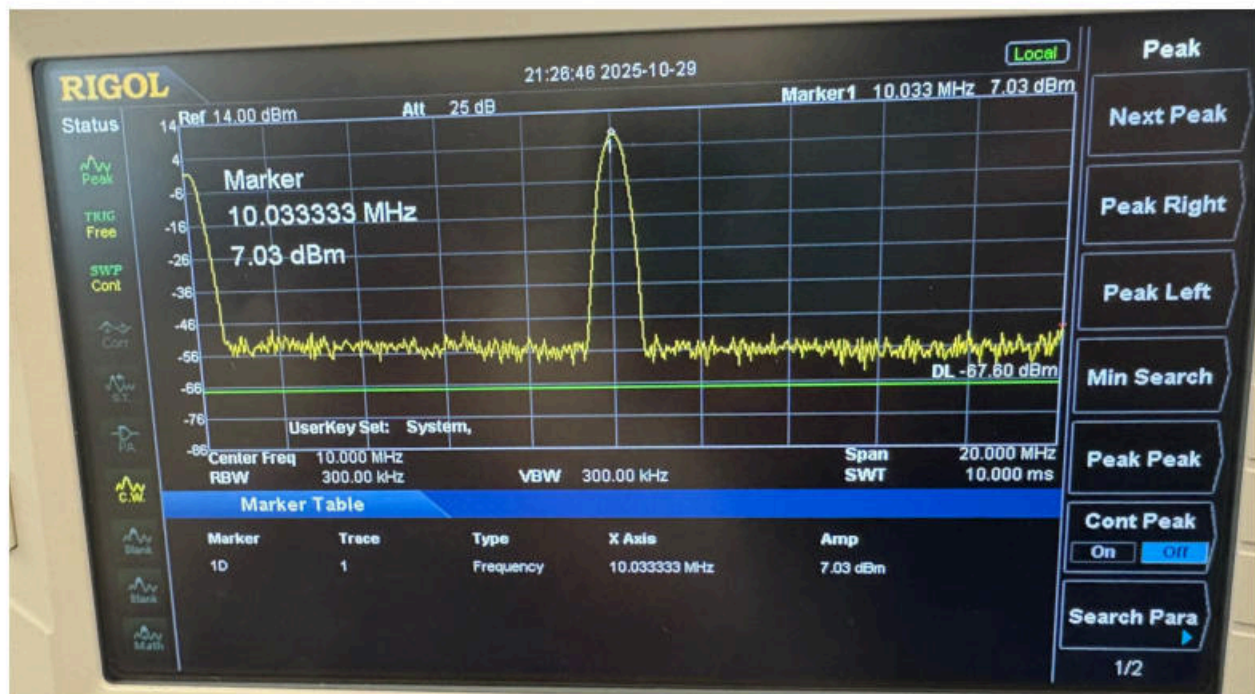
The X stands for the product of the multiplication of the L and R signals. This might also be labeled as IF for Intermediate Frequency in some mixer operations.

IMG 3.

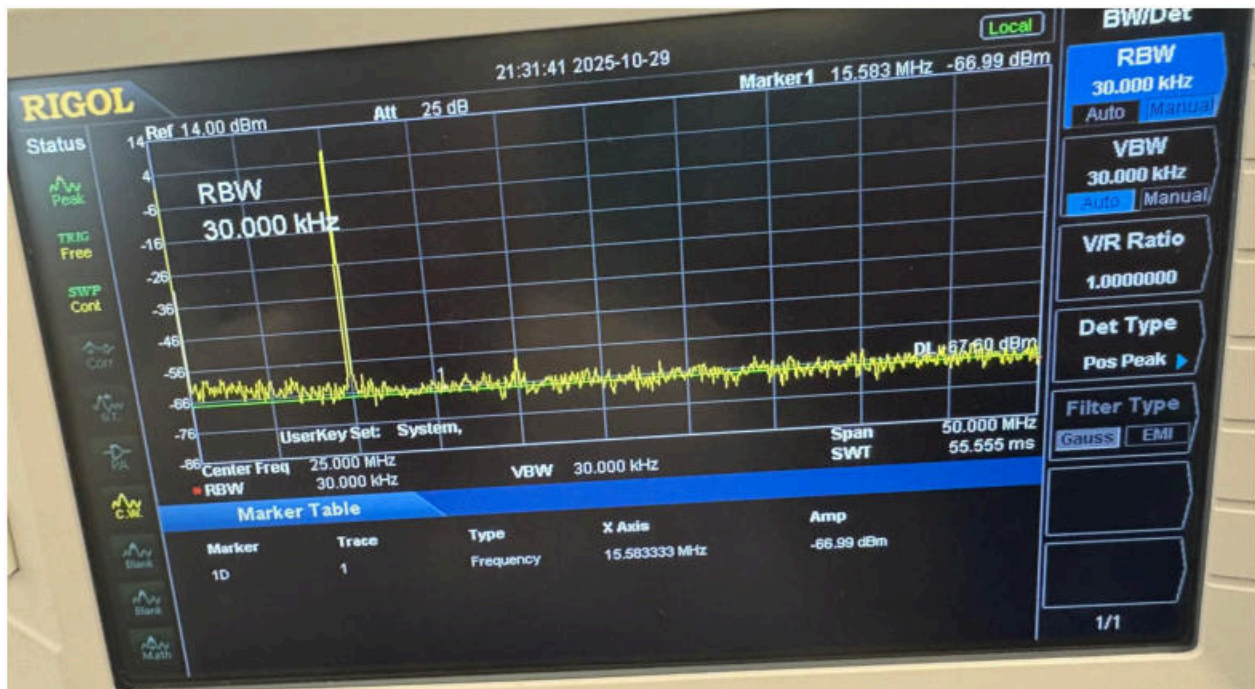
Part A:

1. Setup the Rigol 815DSA Spectrum Analyzer to display frequencies from 0 to 50 MHz.
2. Setup the Rigol 5071 ARB generator for a +7 dBm sinusoid at 10 MHz, start with a signal less than 1 V_{pk-pk} then adjust it while connected to the Spectrum Analyzer. Record the voltage level on the generator once the +7 dBm point is reached, view/record the log (not linear) spectrum making note of the fundamental and any significant frequency terms such as harmonics for comparison with the Agilent generator.

For Part 2, the signal generator was initially set to 500 mV peak-to-peak and connected directly to the spectrum analyzer. The output impedance of the function generator was set to 50 ohms to match the load of the spectrum analyzer, and the output voltage was adjusted accordingly to account for the 50-ohm load setting. We forgot to take a picture of the 50 range, but it looks similar just not centered. ***V_{pp} was shown to equal 1.44V.***



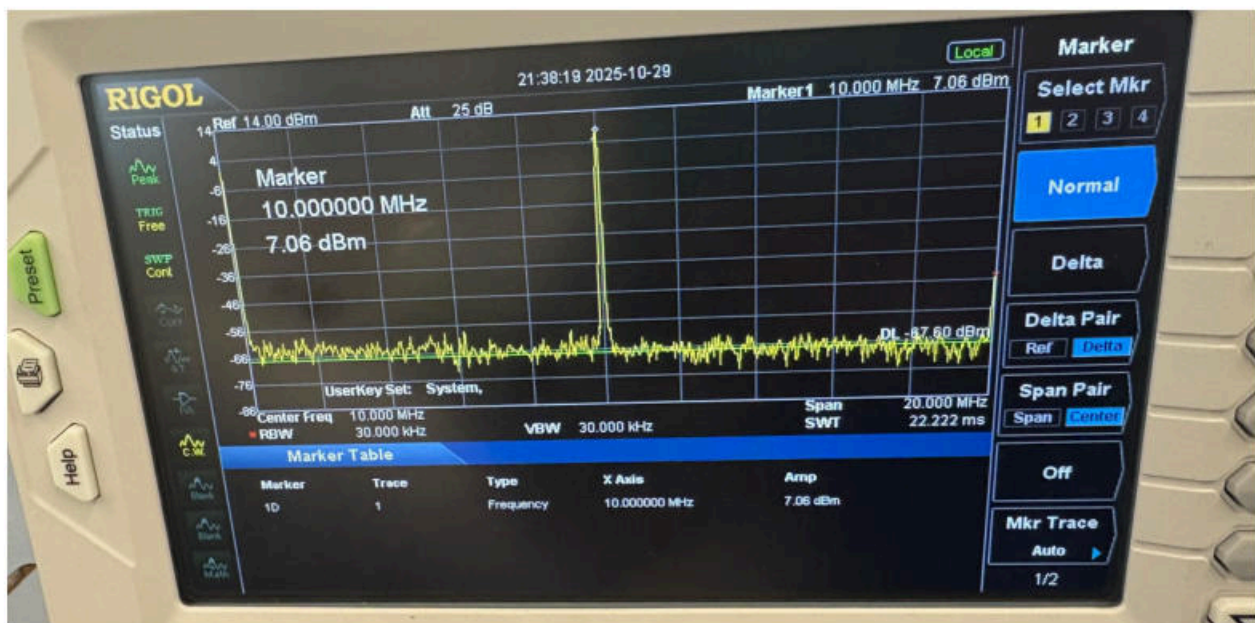
IMG 1.



IMG 2. We just see a second harmonic, as this is a very good generator.

3.Repeat step 2 with the Agilent 33120A ARB generator, _____ Vpk-pk

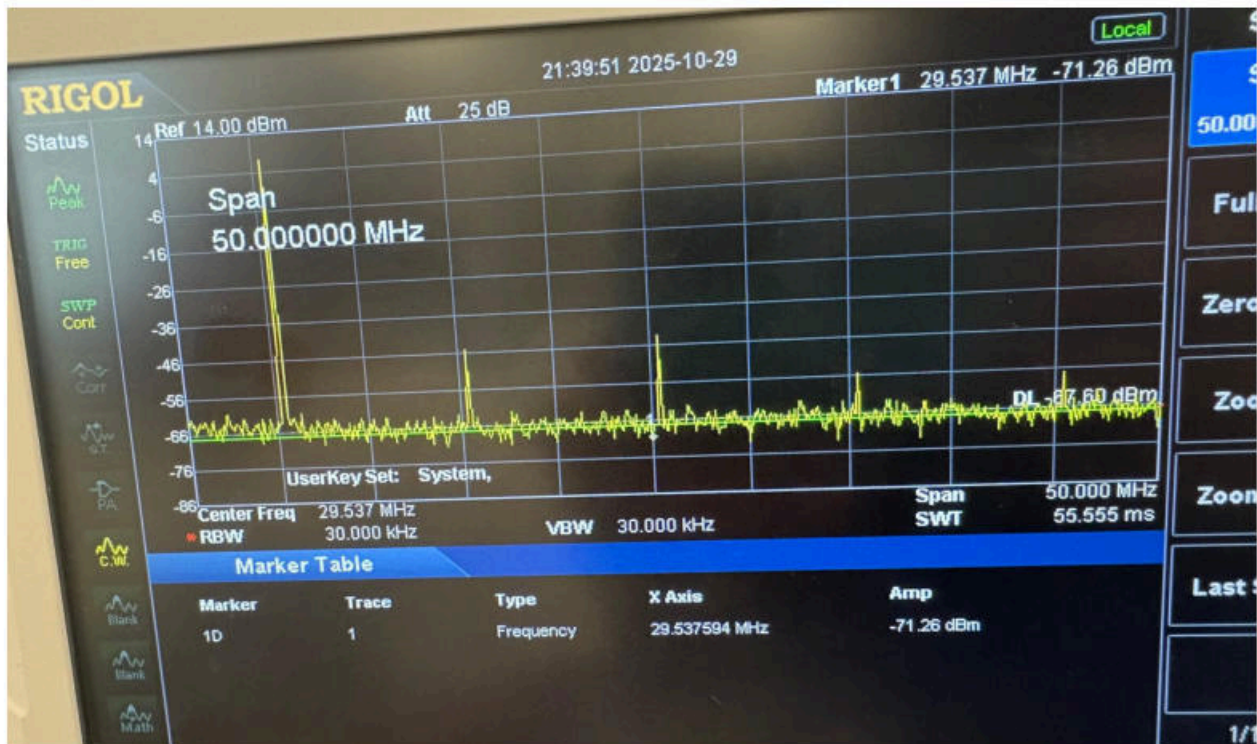
$V_{pp} = 1.53$



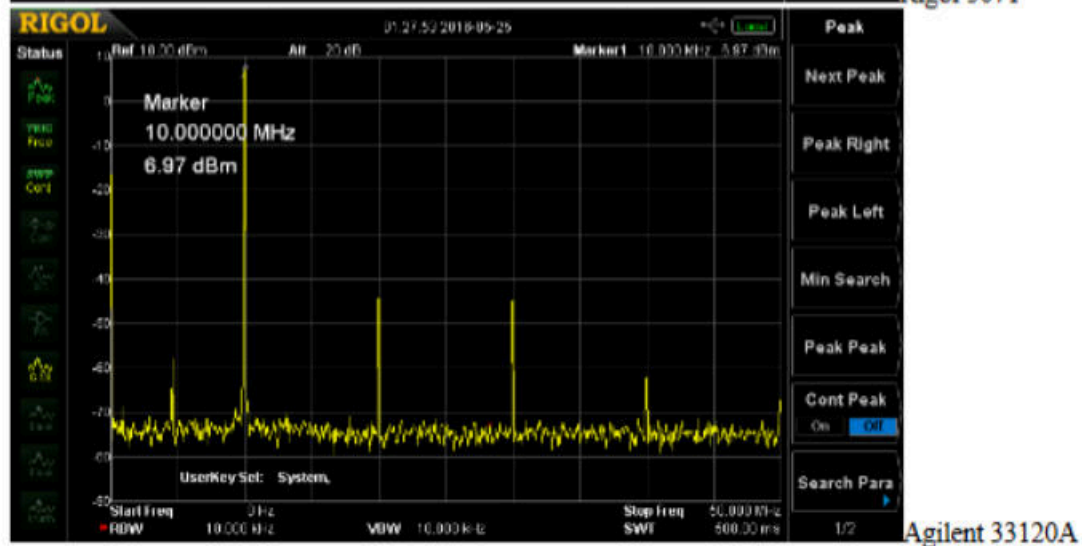
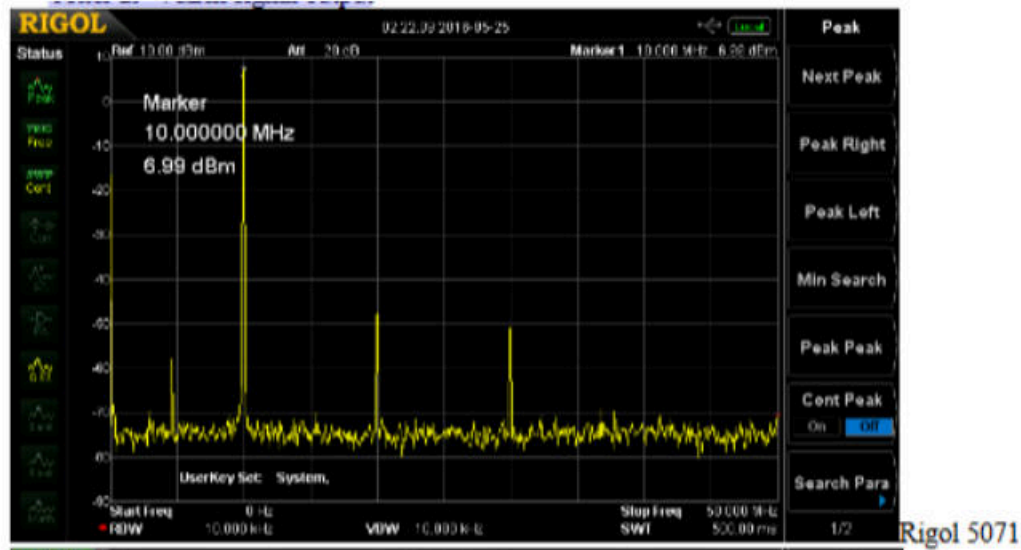
IMG 3.



IMG 4.

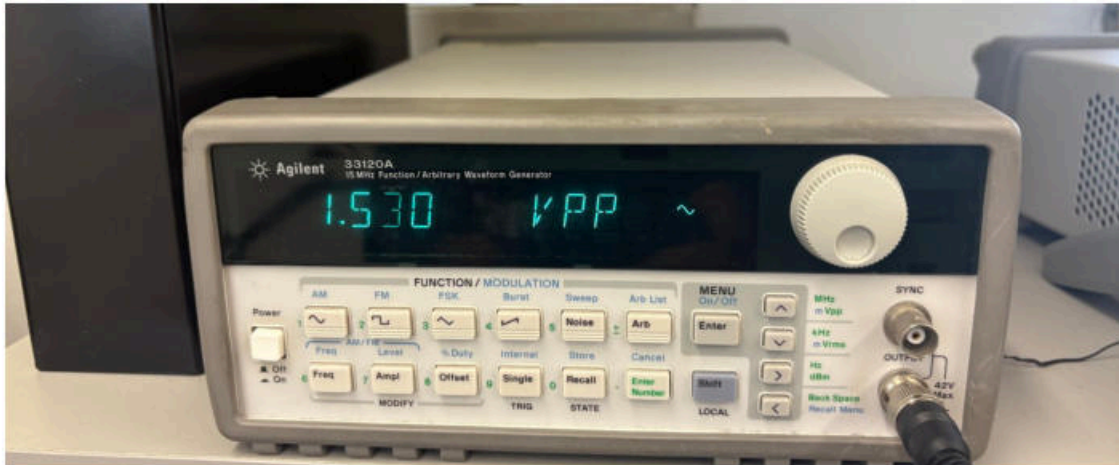


IMG 5.

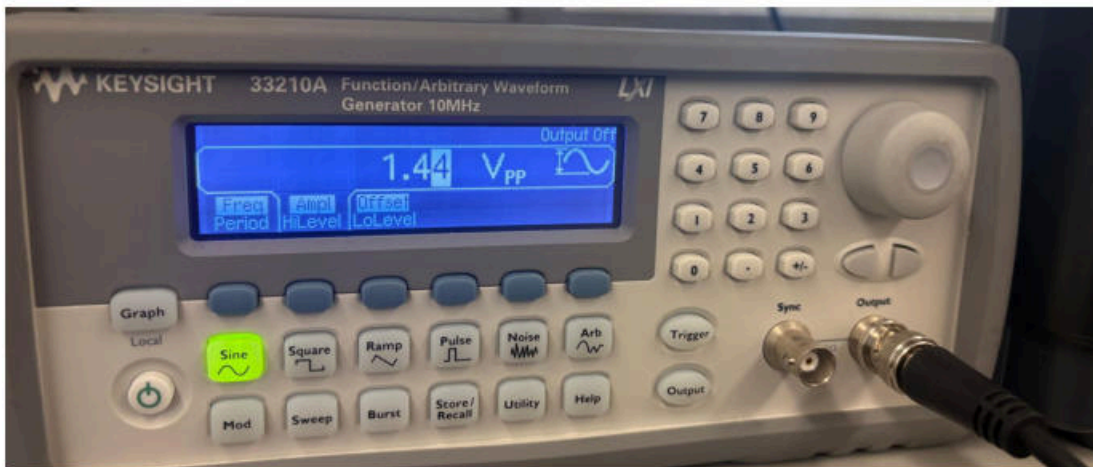


IMG 6. Expected

4. Use the ARB generator with the lowest harmonic content as your LO, in my testing the Rigol was better at +7dBm signal output



IMG 7. R_f



IMG 8. L_o

Part 1.

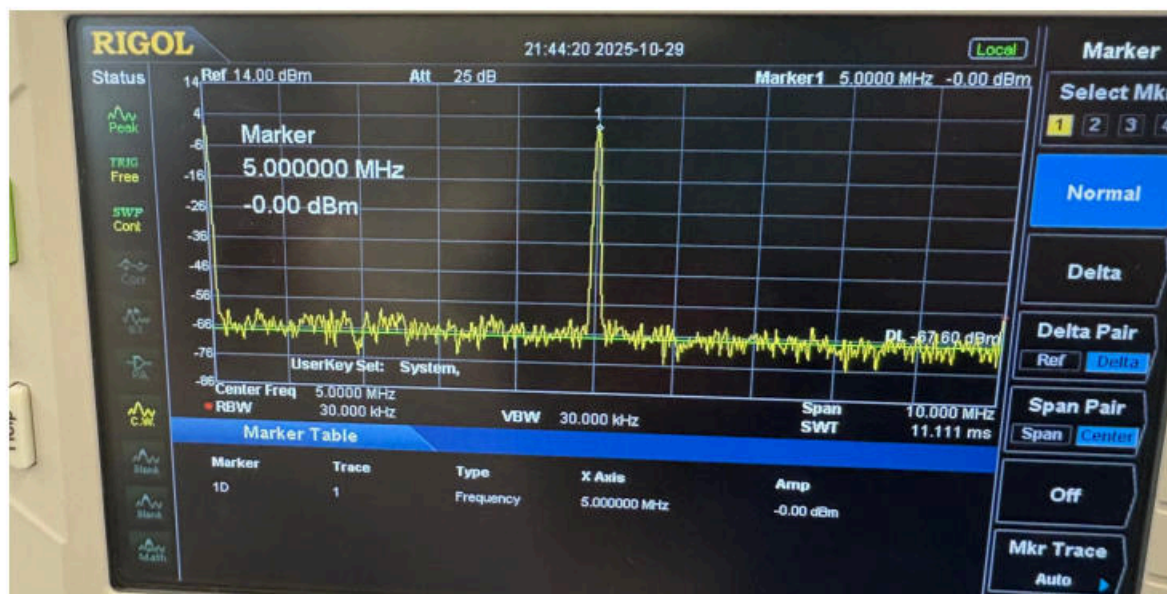
Part 1: Sum and Difference frequency measurements : Fixed LO, variable RF frequency
Signal setup:

1. Setup the Rigol 815DSA Spectrum Analyzer to display frequencies from 0 to 30MHz.
2. Use the LO from part A as a +7dBm sinusoid at 10MHz.
3. Setup the RF generator for a 0dBm signal at 5MHz; record the RF Generator amplitude

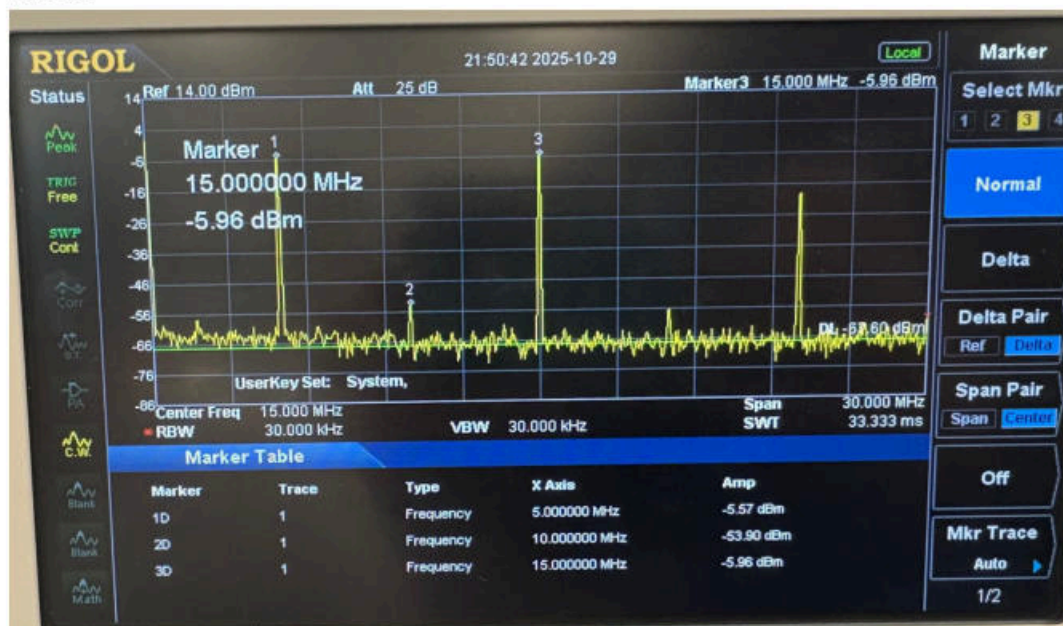
Volts to be able to return to this setting.

The set up steps were completed, and the span was set to 30mhz.
 L_o can be seen in image 8.

3. for 0dbm signal a 5mhz R_f 661.0 mv vpp



IMG 9.



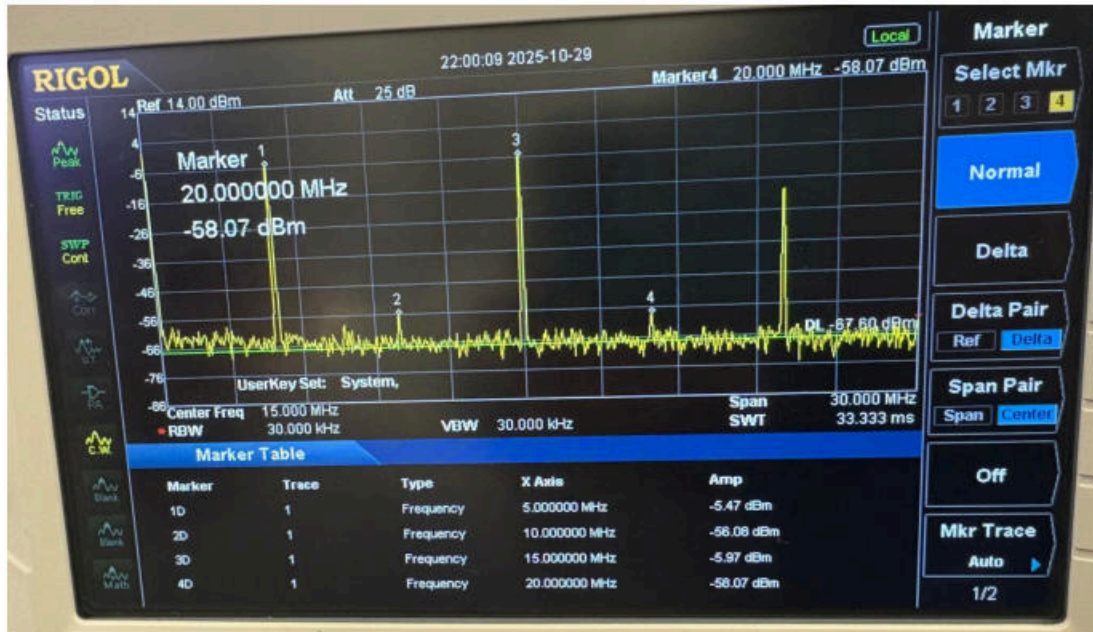
IMG 10.

Connections and Measurements:

4. Connect to the mixer as shown in the figure: LO ARB Generator to the L terminal, RF ARB Generator to the R terminal and finally connect the X terminal to the spectrum analyzer.
5. Confirm that the sum and difference frequencies at 5MHz and 15MHz are displayed, see below. There will also be other signals, but they will typically be at least 20dB below the Sum and Difference signals. In the display below the Peak signal is indicated with a 1 above the yellow signal as Marker 1 at 5.00-MHz and -7.91dBm which is the Difference signal, if PEAK Next Peak is selected Marker 3 at 15MHz would be indicated (RBW or Resolution BW is 30kHz)

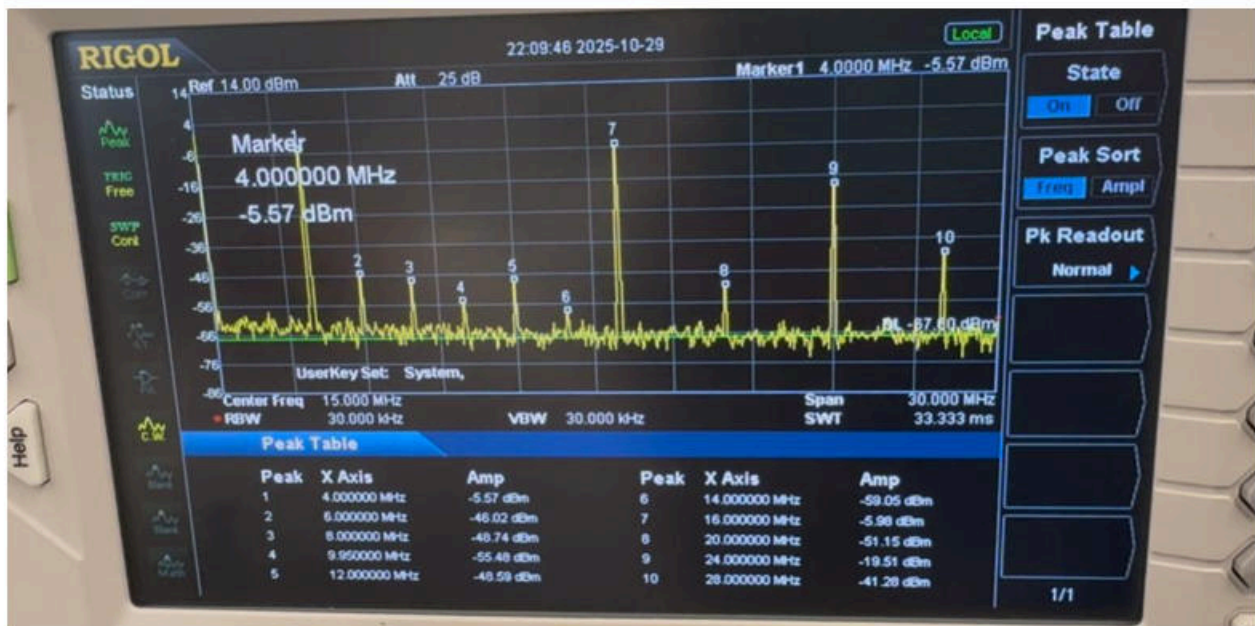
6. As indicated in the above table increase the ARB frequency to 6MHz, then 7MHz up to 15MHz recording the Sum and Diff signal level data for each setting. Note that the Difference signal will be the absolute value of $|LO-RF|$, there are no negative frequencies values in measurements, only in math

For 5mhz...



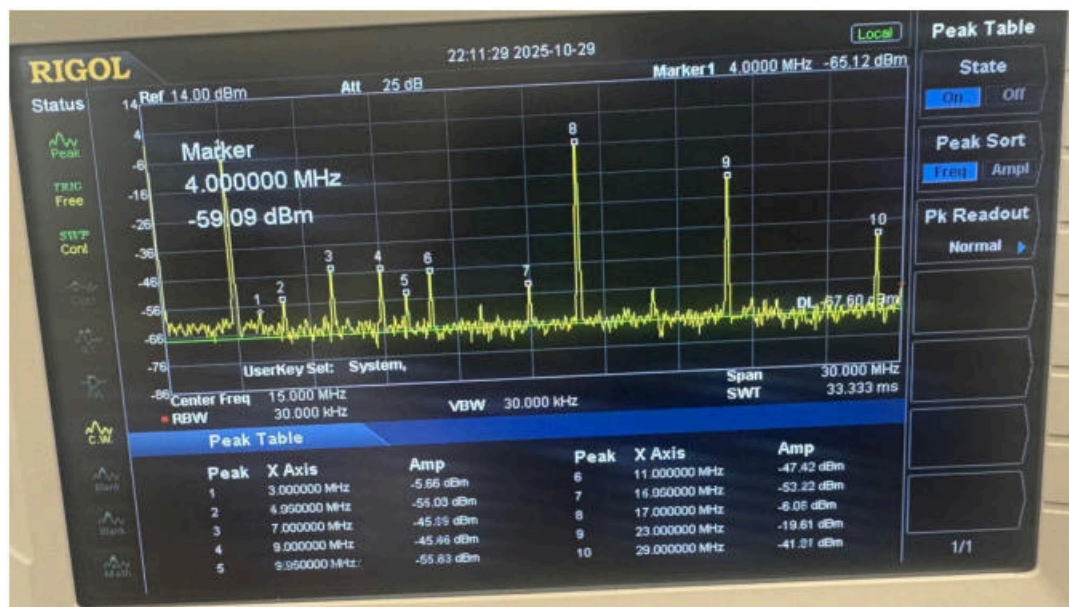
IMG 11.

For 6mhz...



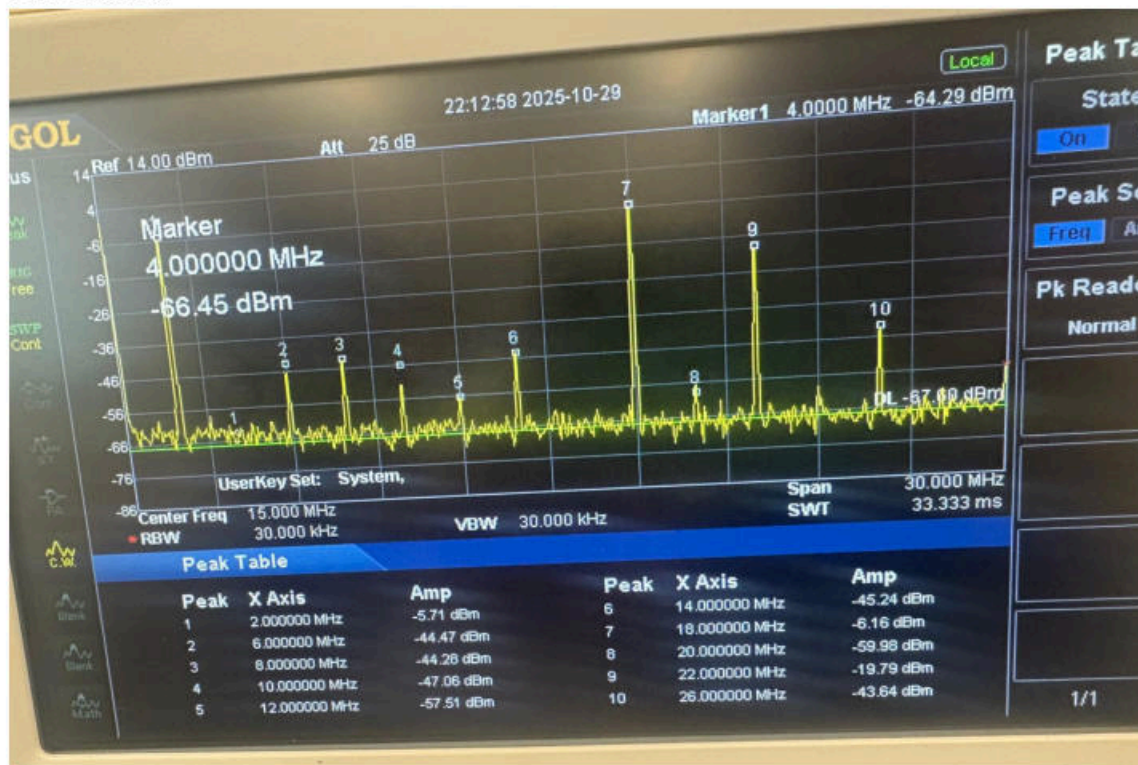
IMG 12.

For 7mhz...



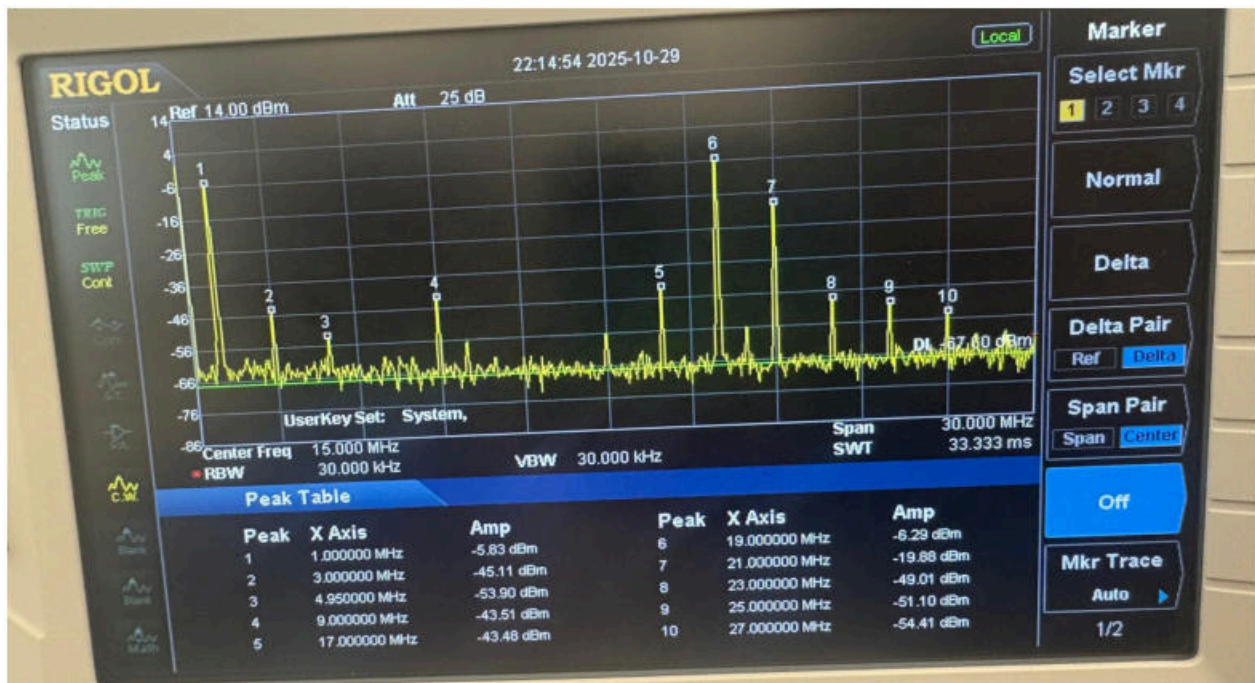
IMG 13

For 8 mhz...



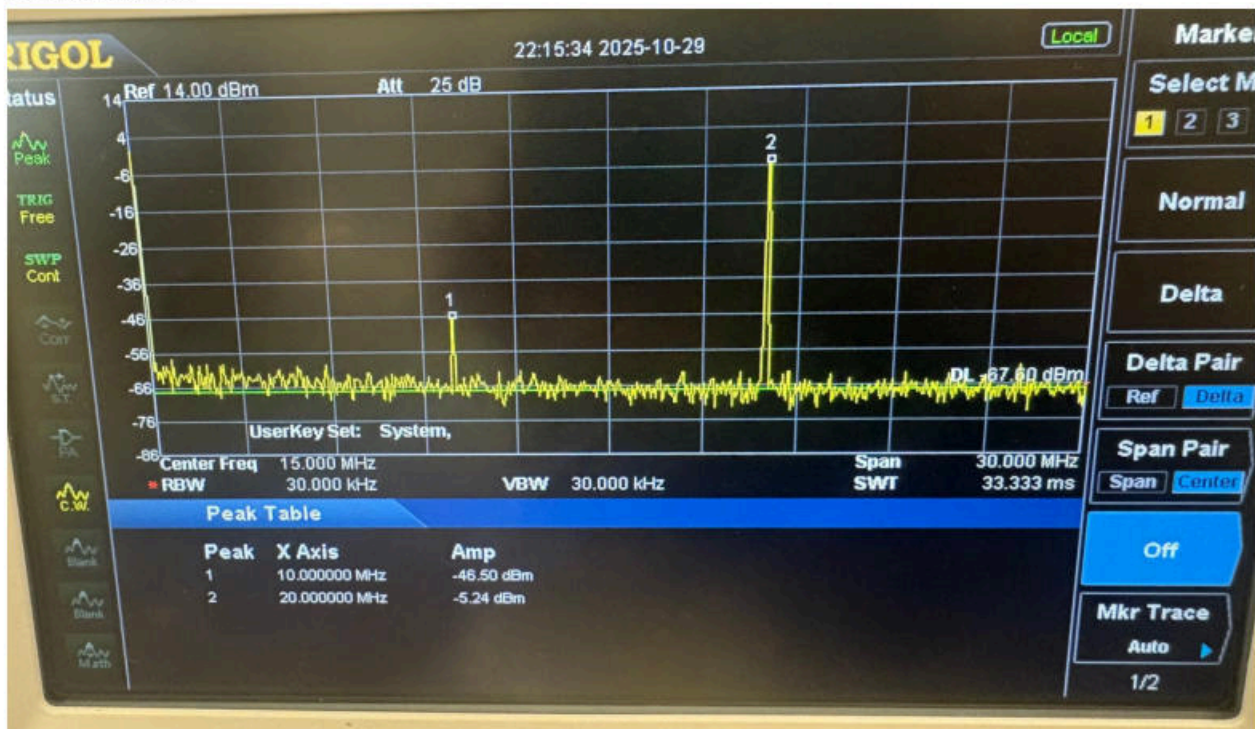
IMG 14.

For 9 mhz...



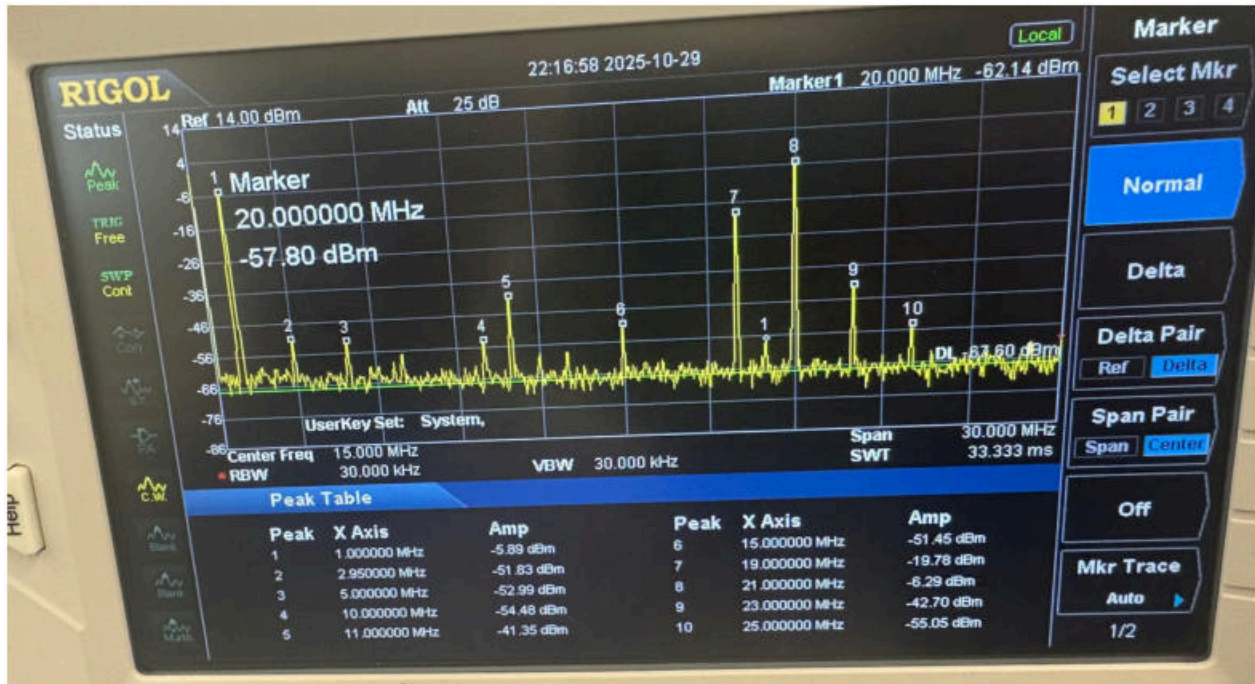
IMG 15.

For 10 mhz...



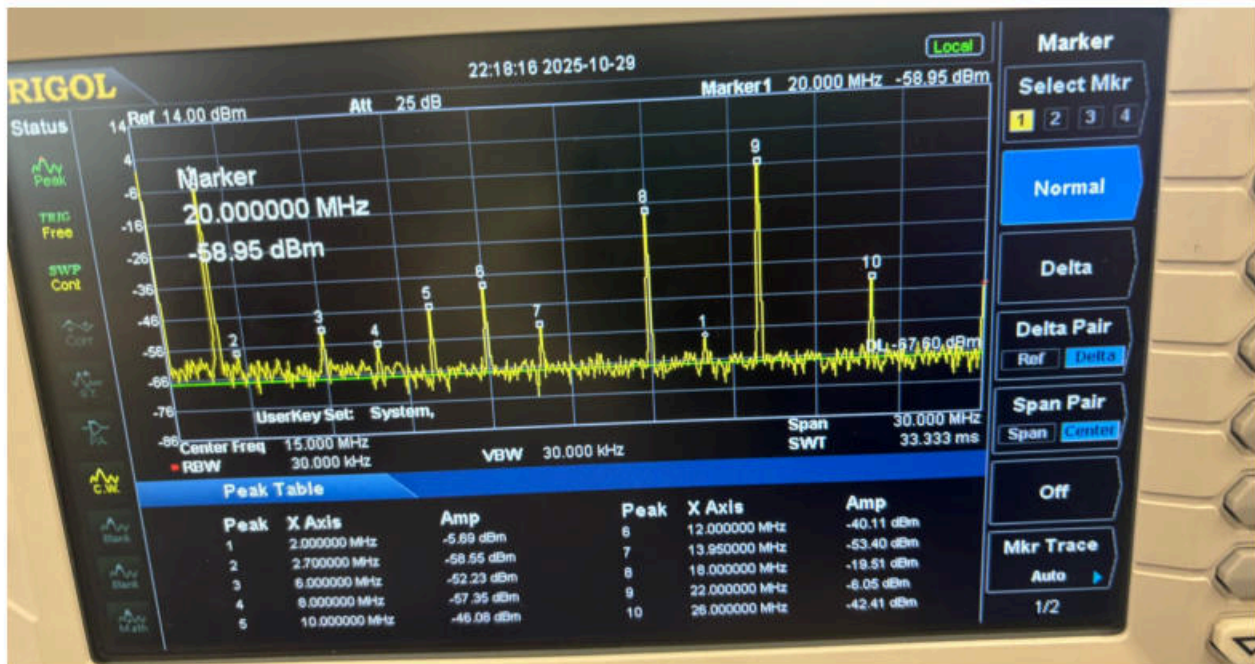
IMG 16.

For 11 mhz...



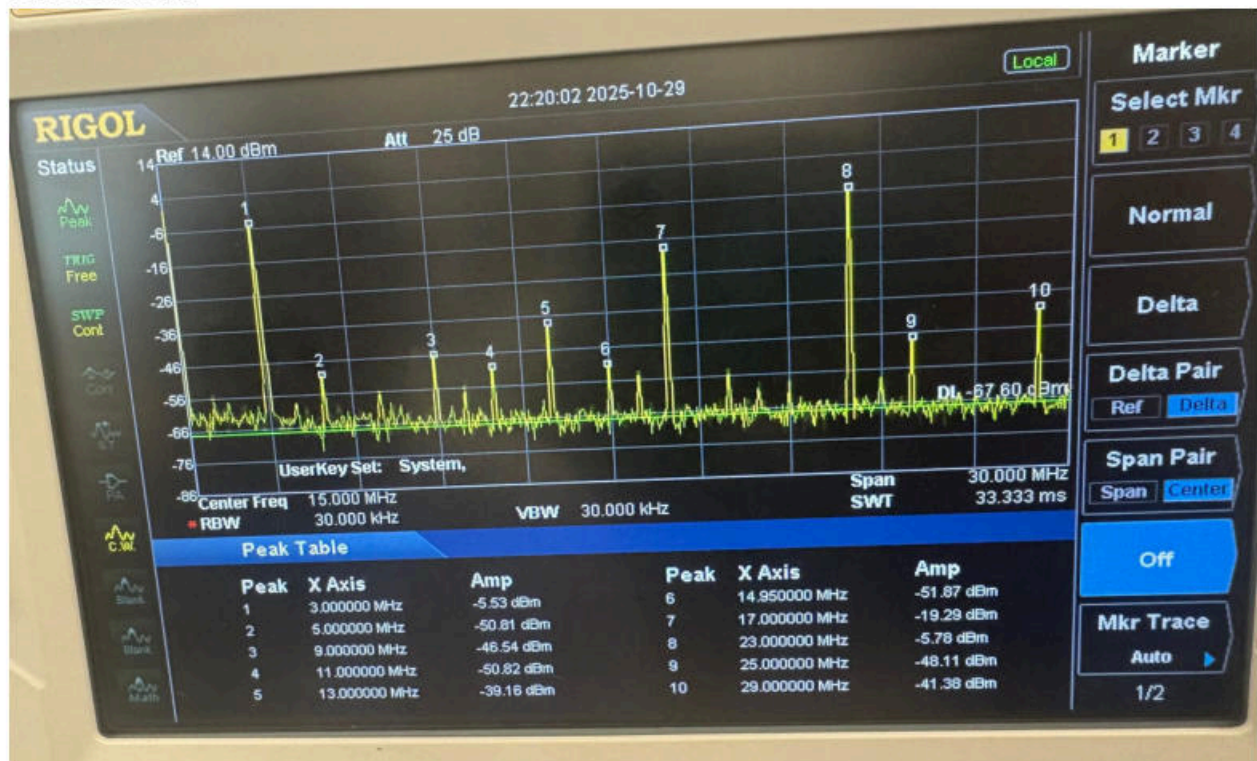
IMG 17.

For 12 mhz...



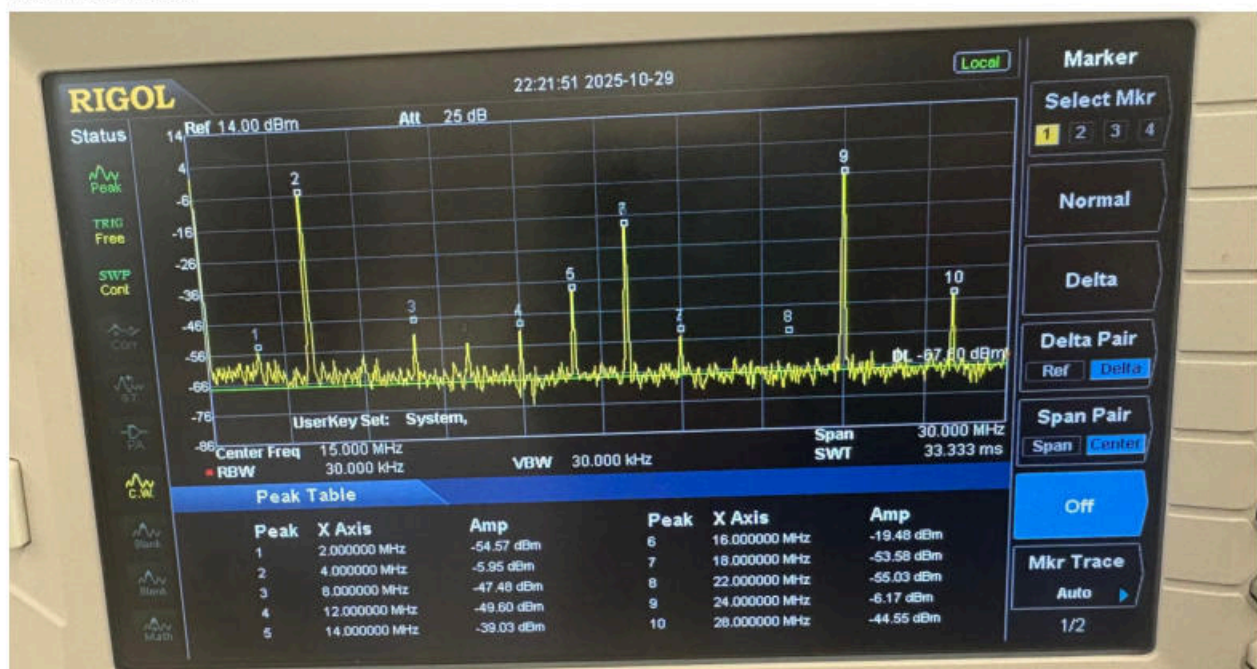
IMG 18.

For 13 mhz...



IMG 19.

For 14 mhz...

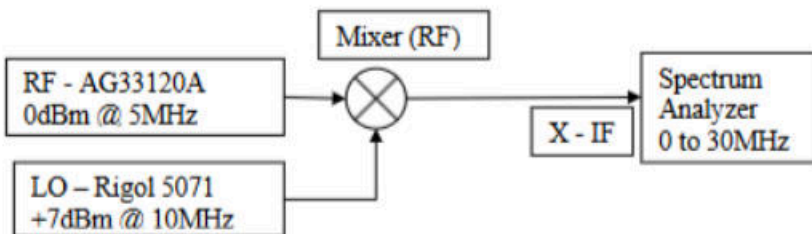


IMG 20.

For 15 mhz...



IMG 21.



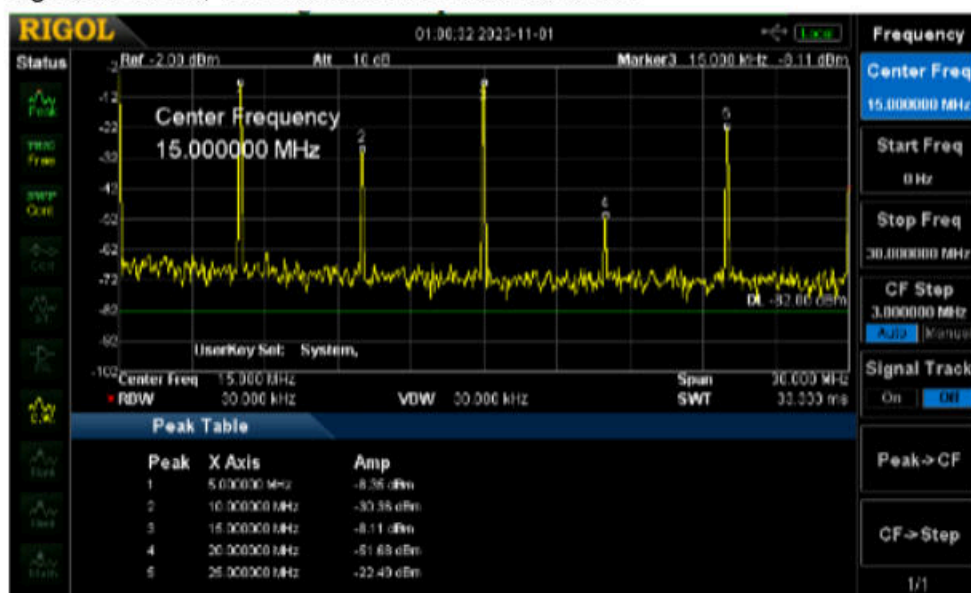
IMG 22.

Lo (Mhz)	RF (mhz)	(Sum) F img= Lo + RF	(Dif) IF = LO -RF	Sum level in dB	Dif level in Db
10	5	15	5	-45.07	-48.47
10	6	16	4	-45.98	-55.87
10	7	17	3	-43.22	-59.08
10	8	18	2	-45.22	-66.45
10	9	19	1	-45.35	-63.06
10	10	20	0	-43.54	-

10	11	21	1	-42.78	-58.86
10	12	22	2	-42.41	-55.63
10	13	23	3	-41.81	-53.08
10	14	24	4	-45.05	-50.80
10	15	25	5	-43.86	-45.42

Table 1.

7. Determine the mixer conversion loss by comparing the Difference signal level to the input RF Signal at 0dBm, the loss should be around 6-8dB

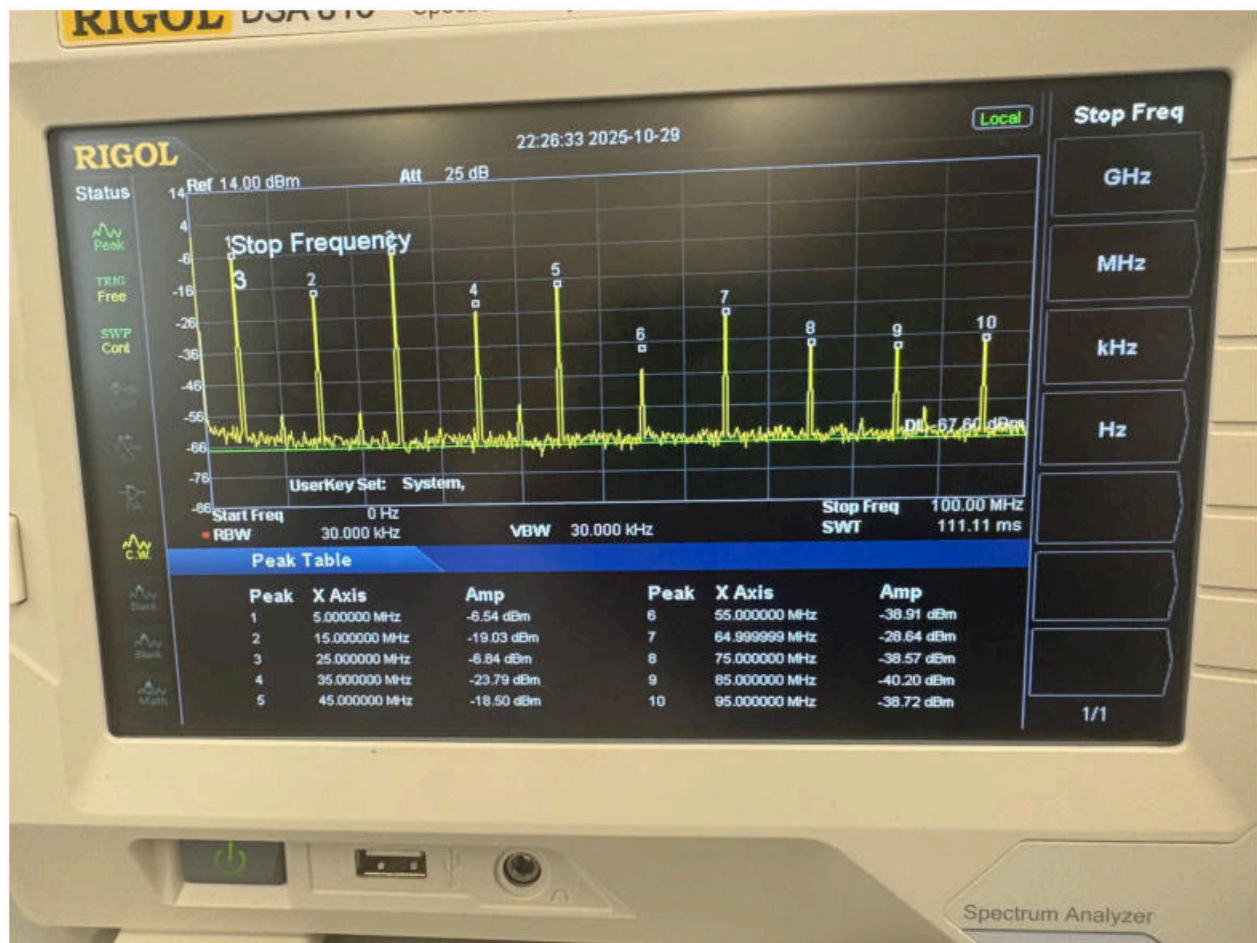


IMG 23.

In the example from class, RF= 0dbm, LO input = +7dbm, and the difference (IF) is around -8 dbm. Therefore, conversion loss = 0 - (-8) = 8 dB.

This loss is roughly constant across the mixer set up. It generally is, except for RF = 10mhz and Rf=15mhz (which is a bit off)

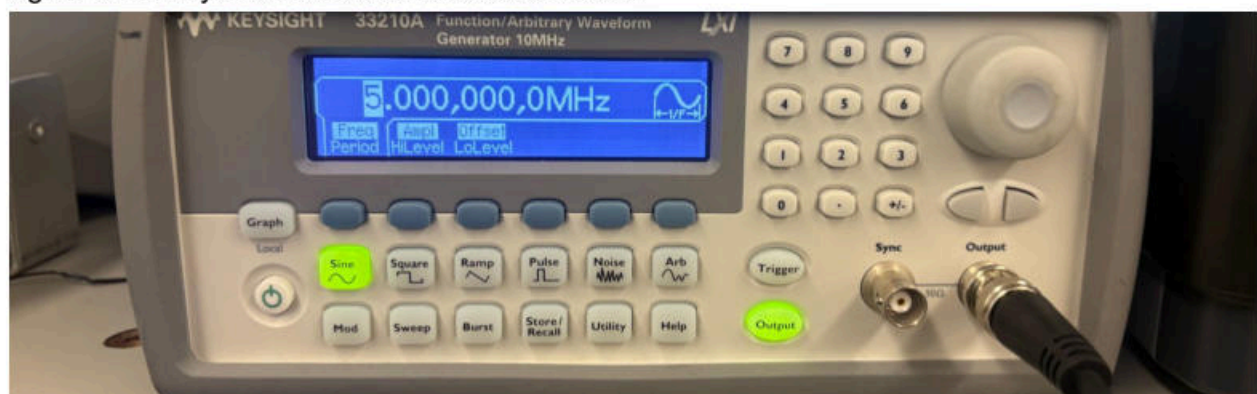
8. To view the full frequency spectrum of the mixing process change the STOP frequency to 100MHz, you will see many frequency terms, this is the downside to the diode mixer used in this lab, it not only creates the very useful Sum and Difference signals it also creates many other frequency terms referred to as intermodulation products which can cause issues if not properly filtered.



IMG 24. Stop frequency changed to 100mhz

Part 2. LO Frequency Variation with a fixed RF frequency (the normal radio receiver setup)

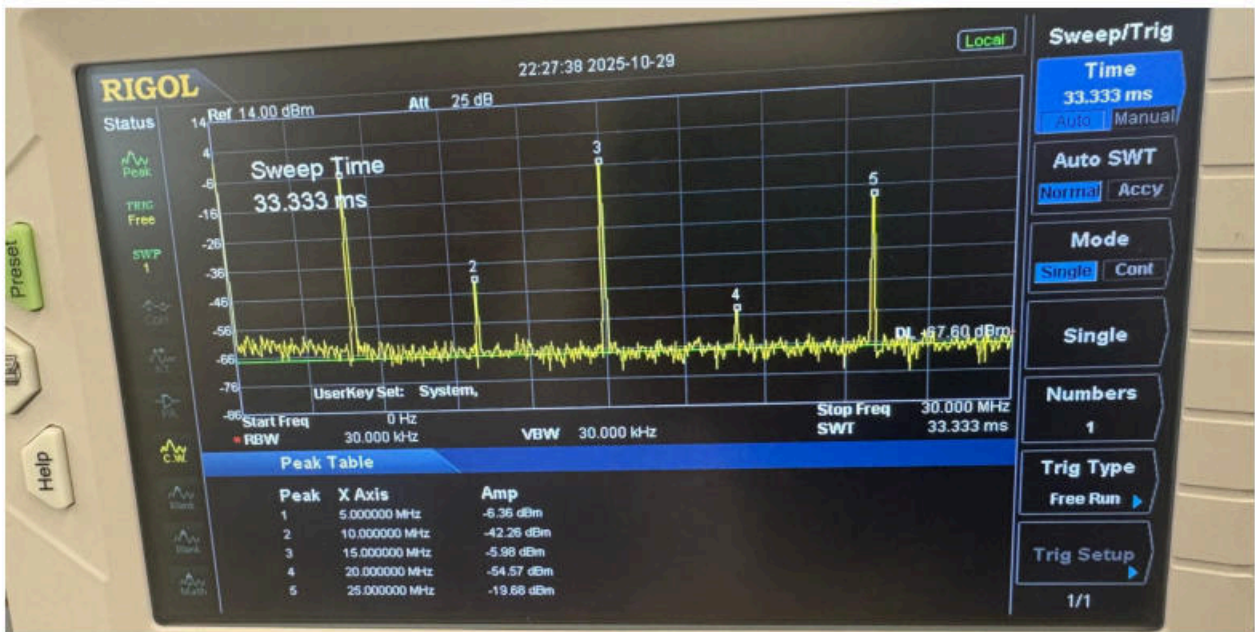
9. Return the system to the part 1 step 4 settings above but make the LO signal 5MHz and the RF 10MHz so the LO is at a lower frequency than the RF signal. Note again the Difference signal will always be the absolute value of $|LO - RF|$



IMG 25. LO

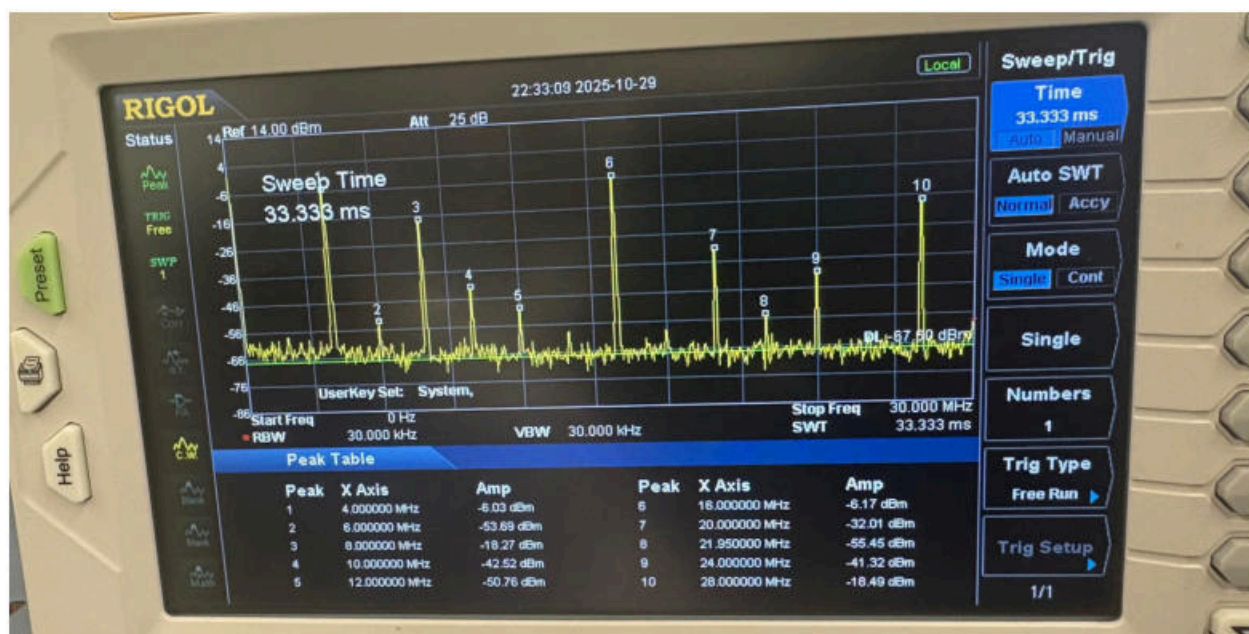


IMG 26. RF

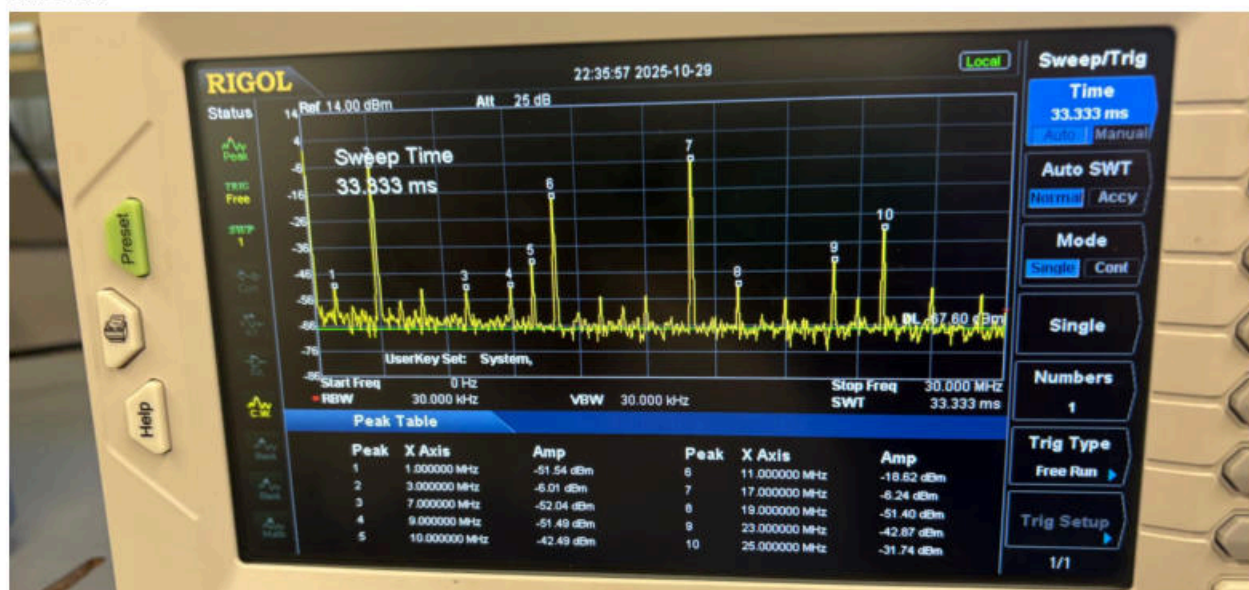


IMG 27.

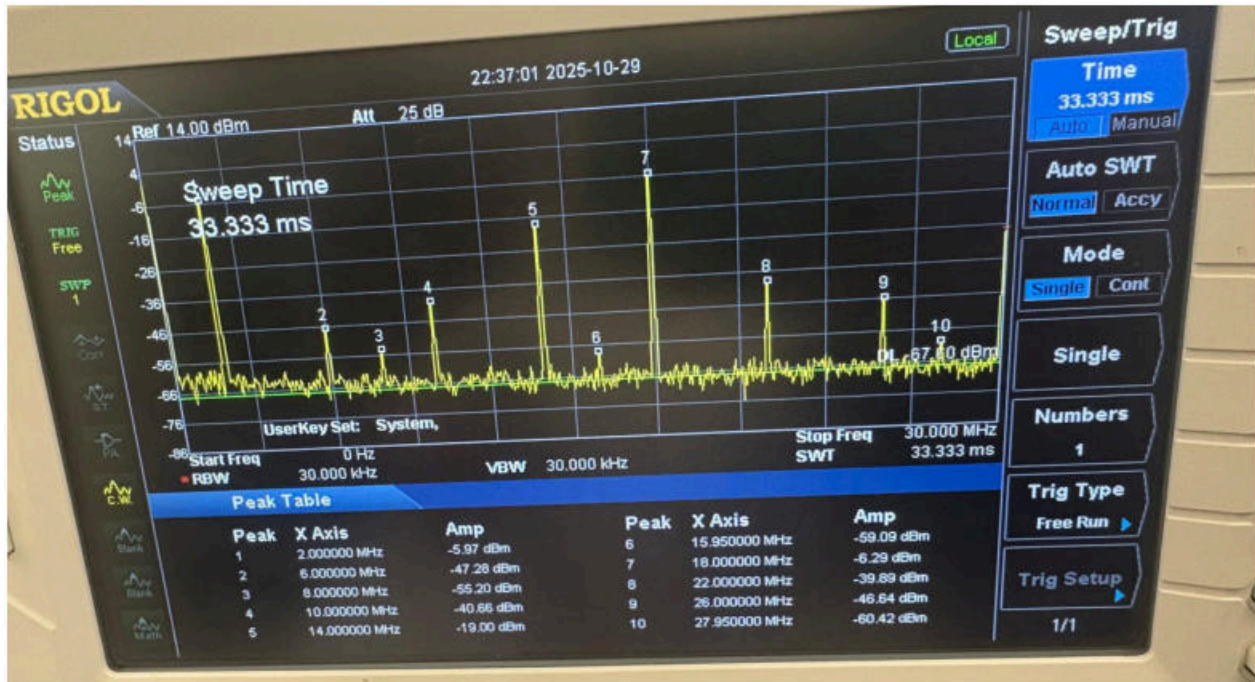
10. Adjust the LO signal frequency up to 15MHz making note of how the signals change, save a sample screenshot for your lab summary



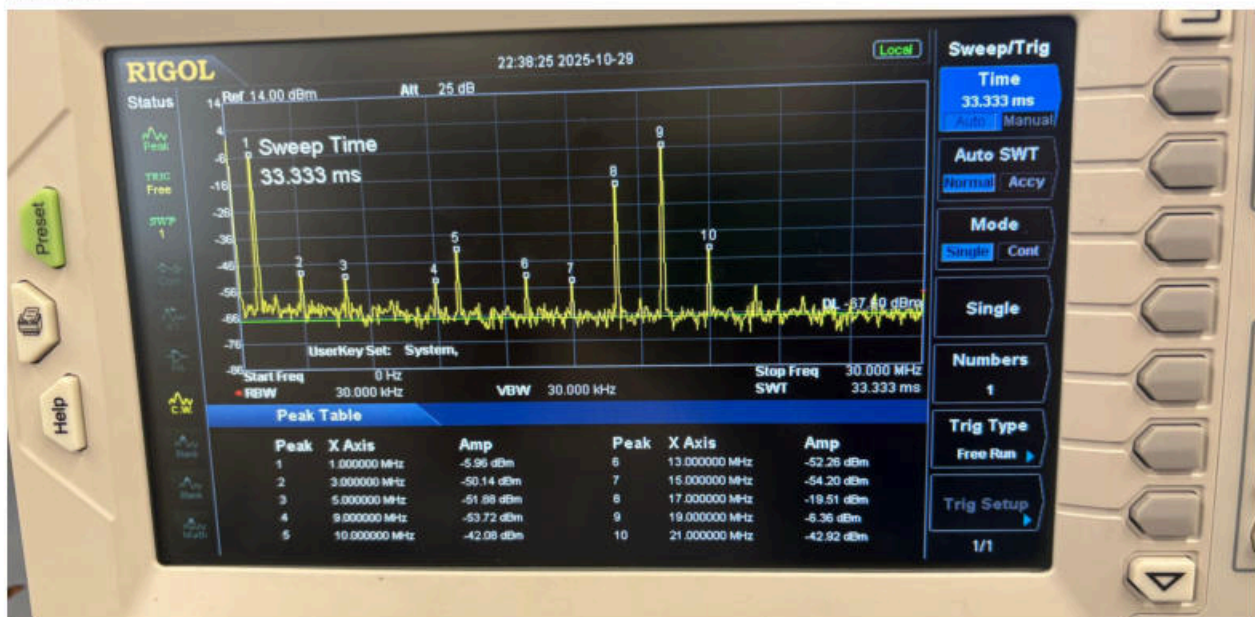
IMG 28.



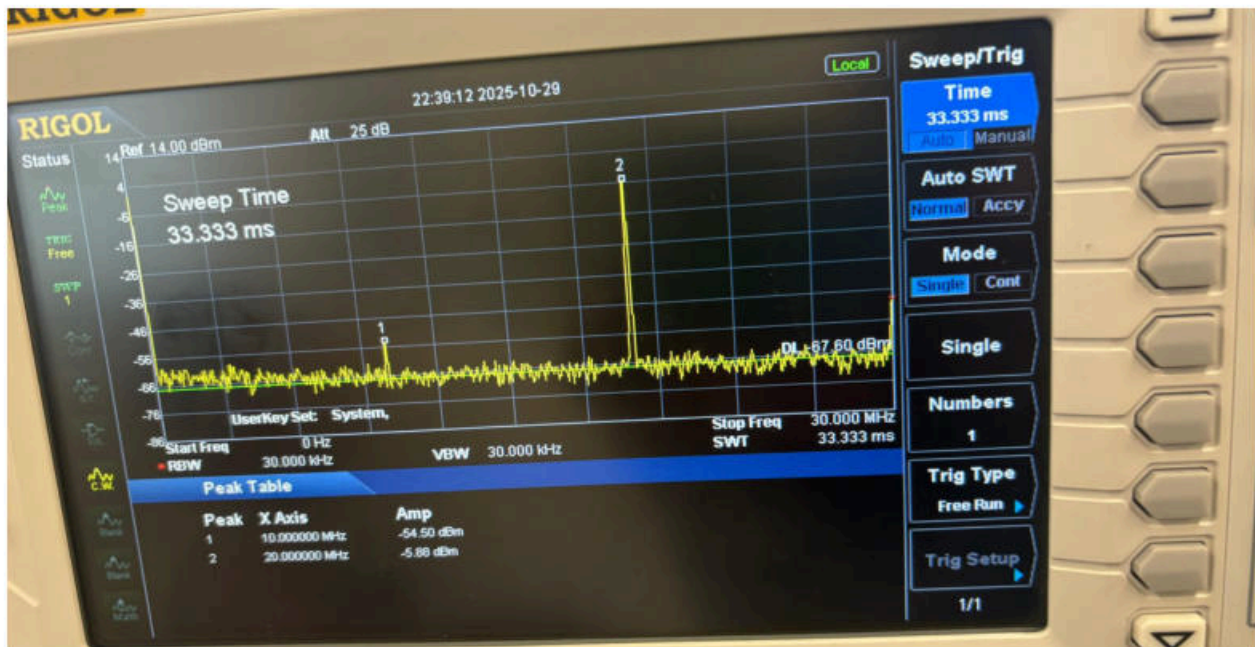
IMG 29



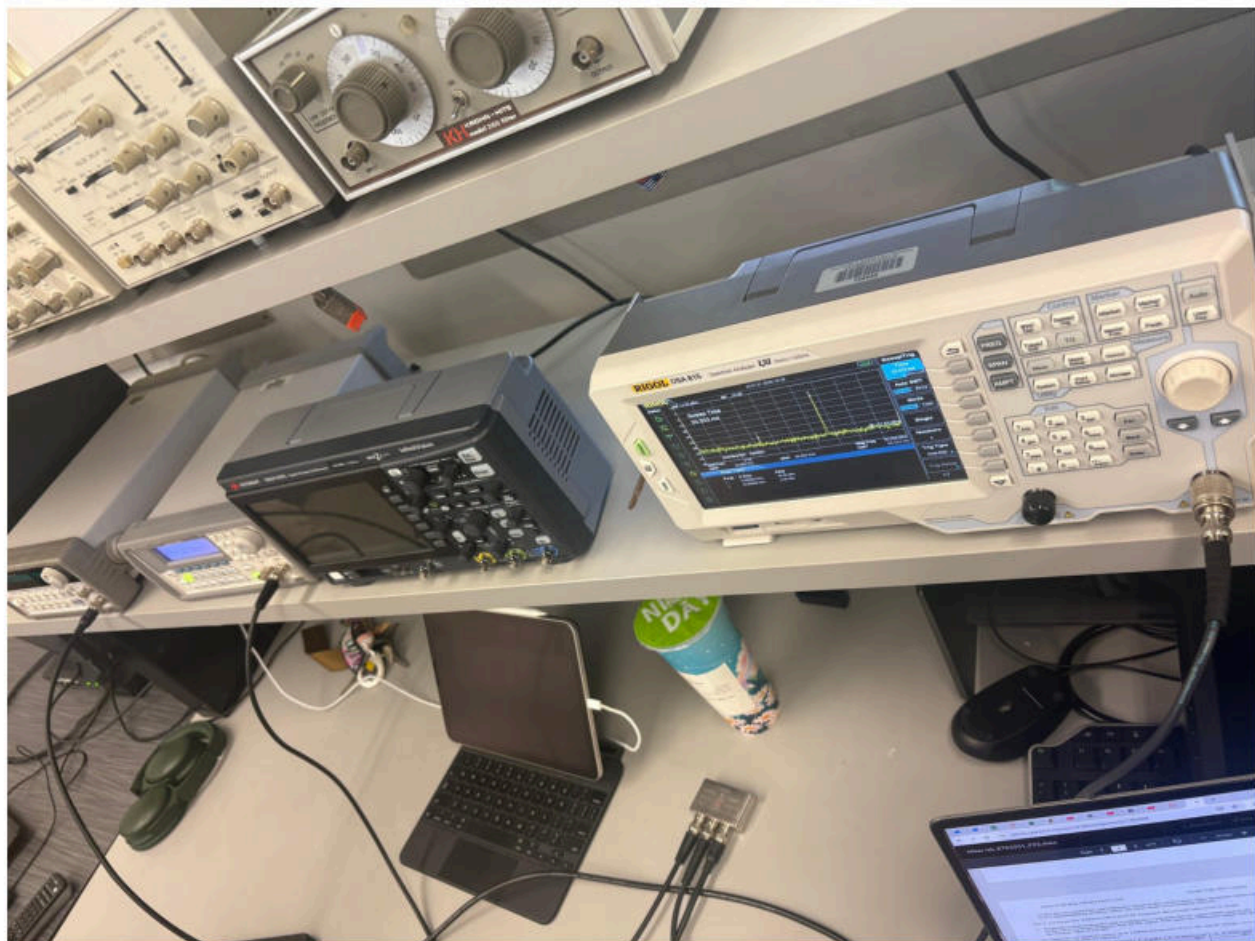
IMG 30



IMG 31.



IMG 32.



IMG 33. Set up

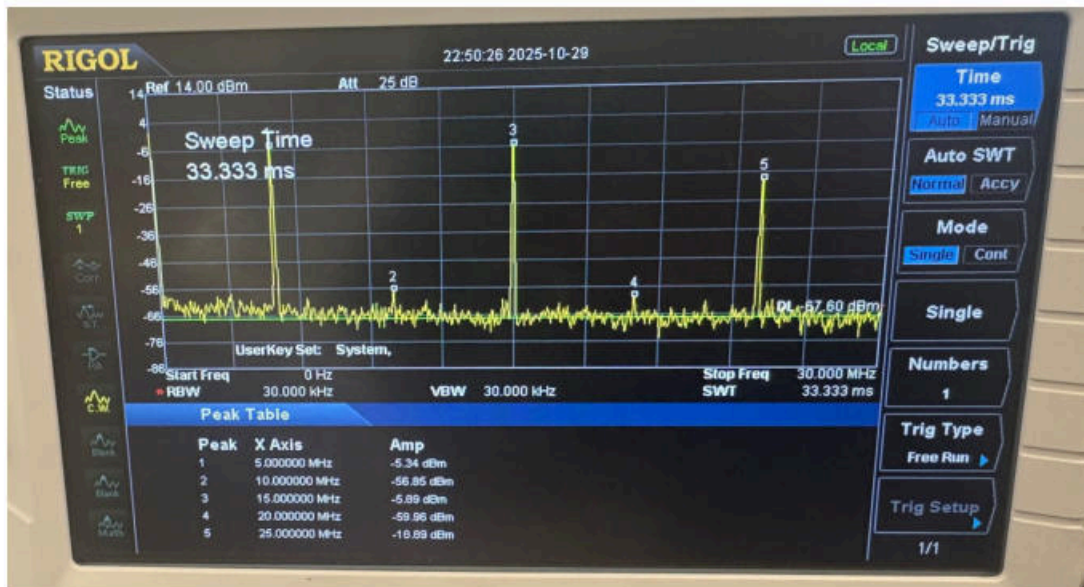
Lo (Mhz)	RF (mhz)	(Sum) F img= Lo + RF	(Dif) IF = LO -RF	Sum level in dB	Dif level in Db
5	10	15	5		
6	10	16	4		
7	10	17	3	-18.61	-50.14
8	10	18	2	-45.22	-66.45
9	10	19	1	-45.35	-45.08
10	10	20	0	-5.88	-
11	10	21	1		
12	10	22	2		
13	10	23	3		
14	10	24	4	-45.05	-50.80
15	10	25	5	-43.86	-45.42

Table 2.

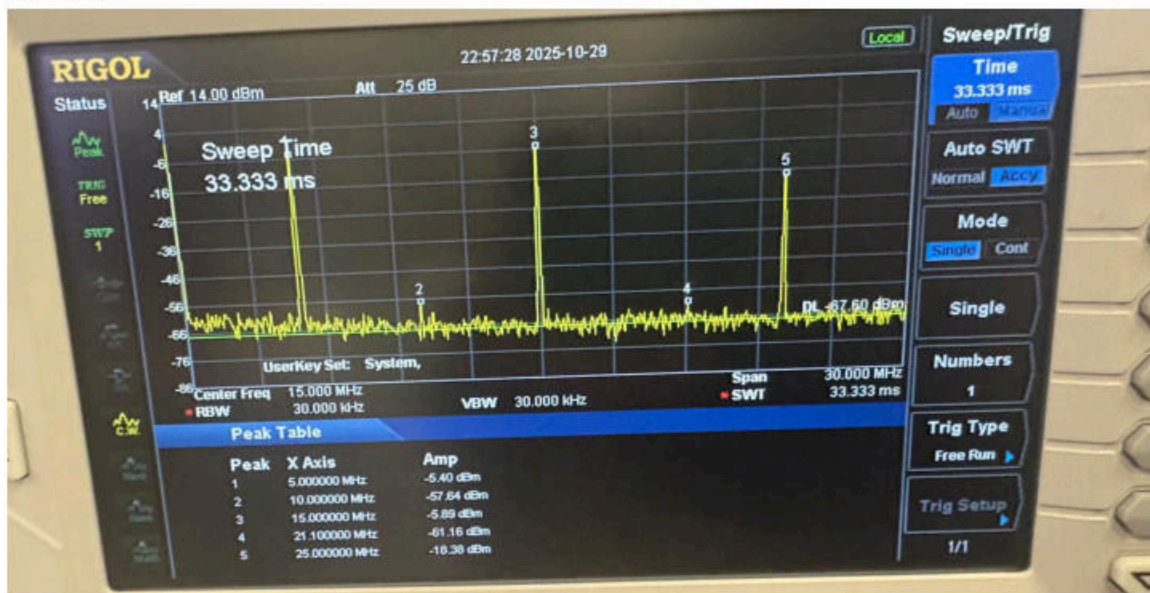
Part 3. RF level variation effects

11. After returning the system to the original part 1 step 4 settings: RF 0dBm signal at 5MHz and the LO +7dBm at 10MHz decrease the RF signal by 10dB to -10dBm (this should be a change in voltage of around $1/\sqrt{10}$ or 0.316) then by -20dB a voltage change of 1/10 or 0.1 from the original settings, recording spectrum changes in a table or screenshots. This step simulates a change in the incoming RF signal from an antenna which occurs regularly in an RF environment, and points out the need for Automatic Gain Control or AGC

In this step the RF signal level was reduced from 0 dBm to -10 dBm and then to -20 dBm while keeping the LO fixed at +7 dBm and 10 MHz. Decreasing the RF power by 10 dB reduced the RF voltage to approximately 0.316 of its original value, and a 20 dB reduction lowered the voltage to about one-tenth of the original level. As the RF amplitude decreased the amplitudes of both the sum and difference frequency components in the mixer output decreased proportionally (which was observed on the spectrum analyzer). This behavior simulates real RF environments where received signal strength varies and demonstrates the need for Automatic Gain Control (AGC) to maintain consistent receiver performance/



IMG 34



IMG 35

Power is proportional to the square of voltage. This means that when the voltage is reduced, the power decreases much faster. A 10 dB decrease in power corresponds to a voltage decrease by a factor of 1 divided by the square root of 10, which is approximately 0.316. A 20 dB decrease in power corresponds to a voltage decrease by a factor of 1 divided by 10. Because power is proportional to voltage squared divided by resistance, reducing the voltage by a factor of 10 reduces the power by a factor of 100.

Power Level (dbm)	Vpp (v)
0	0.63
-10	0.20
-20	0.063

Table 4.

Part Adding modulation to the RF signal

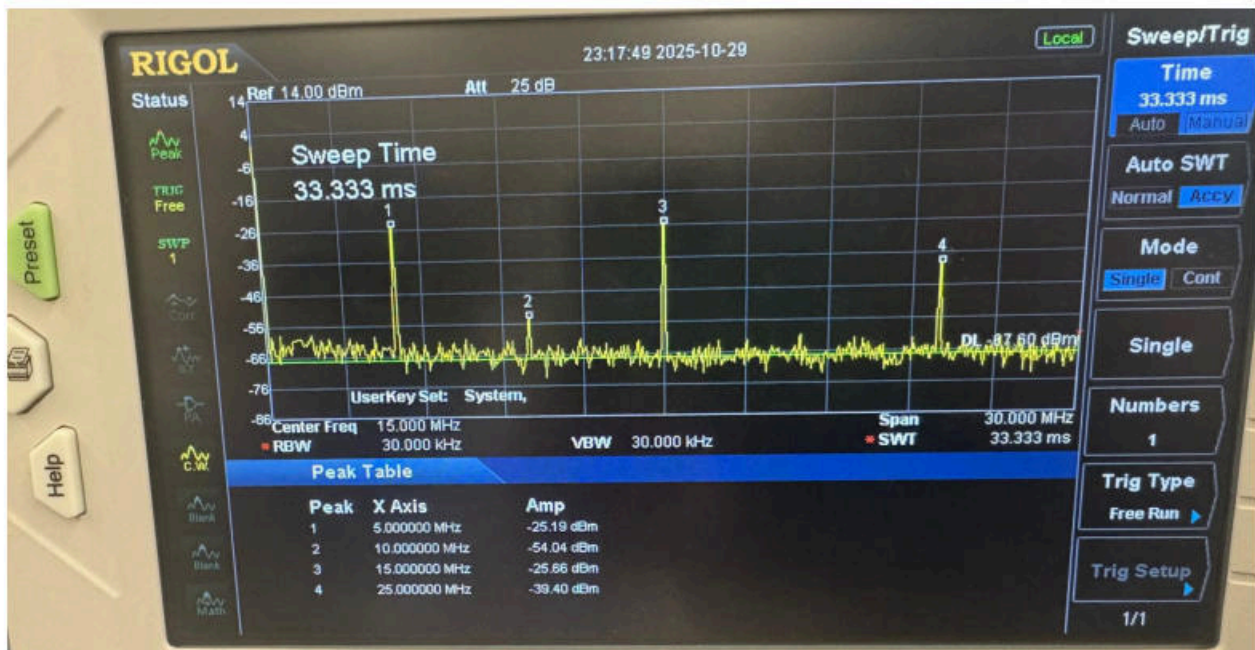
12. Return the RF level to 0dBm and add sinusoidal modulation to the RF signal use an fm value of 10kHz and an ma value of 100%. To view signals near the Difference signal use the FREQUENCY control to put 5MHz in the center of the display and the SPAN control to view only 100kHz, 50kHz on either side of 5MHz, repeat the same for the Sum frequency at 15MHz. You will need to have the Resolution BW set less than 10kHz to views the sidebands. Notice the USB and LSB signals on either side of the sum and difference frequencies as was seen in the AM lab, record screenshots.

For 467 vpp or -10bm:



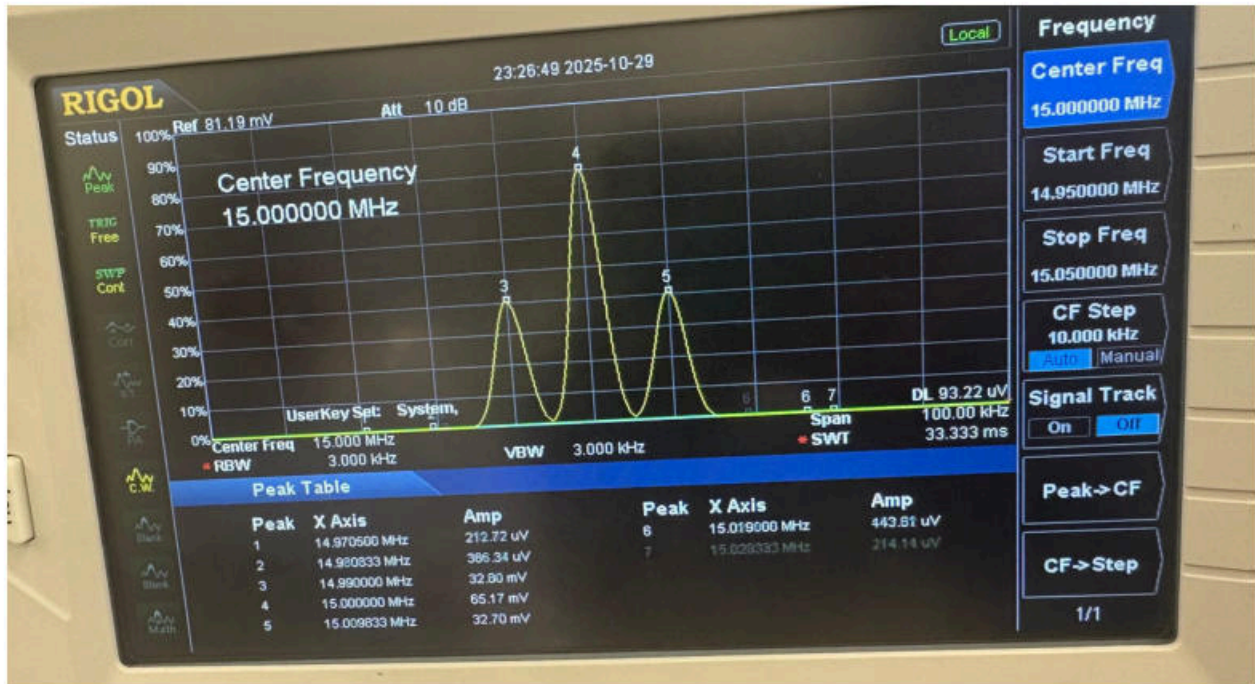
IMG 36

For 66.1mv or -20dbm



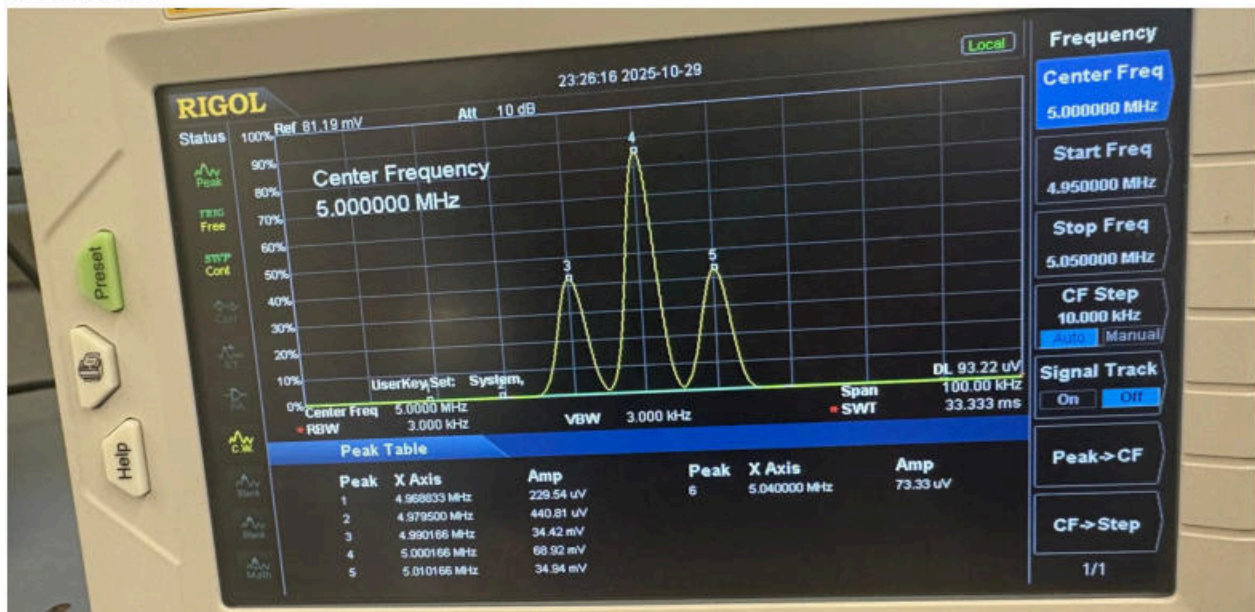
IMH 37

From the mixer output, the spectrum analyzer was centered at 15 MHz and the frequency span was reduced to zoom in on the signal. When the RF signal was set to 5 MHz, clear sidebands appeared around the sum frequency, confirming the expected mixing behavior



IMH 38

Difference



IMG 39

Conclusion

In this lab the operation of an RF mixer was studied by observing how input RF and LO signals combine to produce sum and difference frequency components. By using a fixed Local

Oscillator and varying the RF signal (and later fixing the RF while sweeping the LO) the expected mixer relationships were confirmed. The expected and confirmed resulting relationship was that output frequencies appearing at the sum ($LO + RF$) as well as the absolute value of the difference ($LO - RF$). The spectrum analyzer measurements showed that while frequencies follow predictable equations, the amplitudes of the mixer products depend on conversion loss and frequency response. This explains why output levels are lower than the input signals and often appear as negative dBm values.

The lab also demonstrated important practical effects- including the disappearance of the difference signal when it becomes DC and the presence of unwanted mixer products that must be filtered in real receiver systems. Reducing the RF input power showed that both the sum and difference output levels decrease proportionally, illustrating why Automatic Gain Control (AGC) is necessary in real RF environments where signal strength varies. Overall this lab provided a clear understanding of how mixers perform frequency translation, why filtering and gain control are required, and how theoretical mixer behavior appears in real measurement data.